INCH-POUND

MIL-E-24743
29 NOVEMBER 1991

MILITARY SPECIFICATION

ENCLOSURE, COMMON ELECTRONIC EQUIPMENT,

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the design, manufacture and testing of a water-cooled, common electronic equipment enclosure (CE^3) for shipboard use within sheltered equipment spaces. The CE^3 is intended to accommodate a variety of Standard Hardware Acquisition and Reliability Program (SHARP) hardware elements, including Standard Electronic Modules (SEM) Formats B or D; Standard Power Supplies (SPS) family S1, style 3; and Standard Enclosure System (SES) backplane components, backplane assemblies, and support hardware.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, D.C. 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-S-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for.
MIL-C-5015	-	Connectors, Electrical, Circular Threaded, AN Type, General Specification for.
MIL-W-21965	-	Water Cooling of Shipboard Electronic Equipment, General Specification for.
MIL-A-24741	-	Assemblies, Metal Electrical Backplane, One and Two Layer, General Specification for.
MIL-P-24764	-	Power Supplies, Shipboard, Electronic, General Specification for.
MIL-C-28754	-	Connectors, Electrical, Modular, and Component Parts, General Specification for.
MIL-M-28787	-	Modules, Standard Electronic, General Specification for.
MIL-C-28840	-	Connectors, Electrical, Circular Threaded, High Density, High Shock, Shipboard, Class D, General Specification for.
MIL-C-28859	-	Connector Component Parts, Electrical Backplane, Printed-Wiring, General Specification for.
MIL-A-28870	-	Assemblies, Electrical Backplane, Printed-Wiring, General Specification for.

STANDARDS

DOD-STD-100	-	Engineering Drawing Practices.

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MIL-STD-108	-	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-461	-	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-740-1	-	Airborne Sound Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-740-2	-	Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines.
MIL-STD-1310	-	Shipboard Bonding, Grounding, and Other Techniques for Electromagnetic Compatibility and Safety.
MIL-STD-1378	-	Requirements for Employing Standard Electronic Modules.
MIL-STD-1389	-	Design Requirements for Standard Electronic Modules.
MIL-STD-1399 SECTION 300	-	Interface Standard for Shipboard Systems Section 300 Electric Power, Alternating Current (Metric)
DOD-STD-1399 SECTION 532	-	Interface Standard for Shipboard Systems Section 532 Cooling Water for Support of Electronic Equipment (Metric)
MIL-STD-2036	-	General Requirements for Electronic Equipment Specifications
MIL-STD-2038	-	Requirements for Employing Standard Power Supplies
MIL-STD-2073-1	-	DOD Materiel Procedures for Development and Application of Packaging Requirements.
MIL-STD-2119	-	Design Requirements for Printed-Wiring Electrical Backplane Assemblies.
MIL-STD-2198	-	Design Requirements for Metal Electrical Backplane Assemblies.

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MIL-STD-2200 - Requirements for Employing Standard Enclosure Systems.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

Naval Sea Systems Command (NAVSEA)

53711-5942149 - Drawer Assembly, SEM-D

53711-5942190 - Cabinet Assembly, Standard 9 Drawer.

53711-5942198 - Fitting, Supply/Return.

(Copies of the NAVSEA drawings are available from the Naval Weapons Support Center, (ATTN: Code 6042), Crane, IN 47522-5060.)

Naval Underwater Systems Center (NUSC)

88594-01257 - Simulated 25-30 CPS Deck for Shock Testing

(Copies of the NUSC drawing are available from the Naval Underwater Systems Center, (ATTN: Code 4112), New London, CT 06320.)

PUBLICATIONS

Naval Avionics Center

TP-532 - Standard Hardware Acquisition and Reliability Program (SHARP) Hardware Catalog.

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(Copies of TP-532 are available from the Naval Avionics Center, (ATTN: Code 814), 6000 E. 21st Street, Indianapolis, IN 46219-2189.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM F 1166	-	Standard Practice for Human Engineering Design for Marine Systems, Equipment, and Facilities.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 CE^3 baseline configuration. Figure 1 shows a typical enclosure assembly and figure 2 depicts the enclosure's major components.

3.1.1 Standard 9-drawer cabinet assembly. NAVSEA drawing 53711-5942190 shall be the baseline for the cabinet assembly of the CE^3. The cabinet assembly consists of a front structure, a rear structure, a water manifold assembly, and drawer slide assemblies.

3.1.1.1 Front structure. The front structure shall provide the front panel openings necessary to accommodate 9 SEM drawers. The dimensions of the front panel openings shall not exceed 6.7 inches (170.2 millimeters (mm)) wide by 22.75 inches (577.9 mm) high. The forward end of the drawer slide assemblies shall mount to the front structure.

3.1.1.2 Rear structure. The rear structure shall contain the water manifold assembly. The aft end of the drawer slide assemblies shall mount to the rear structure. The rear structure shall have provisions for attachment of the rear connector panel.

3.1.1.3 Water manifold assembly. The water manifold assembly shall be provided within the rear of the enclosure in accordance with MIL-W-21965. The water manifold assembly shall distribute water from the ship's Electronic and Auxiliary Fresh Water (EAFW) system to the SEM drawers. To facilitate replacement of individual SEM drawers, the manifold assembly shall provide for complete isolation of each drawer via positive closing valves located within the enclosure.

3.1.1.4 SEM drawer slide assemblies. The SEM drawer to SEM drawer slide interface shall be designed to allow easy removal of the SEM drawers from the CE^3. Additionally the SEM drawer slide assemblies shall be capable of quick and easy removal from the CE^3 in order to facilitate rapid disassembly/reassembly of the CE^3. See maintainability requirements of 3.17.1.3 and 3.17.1.4.

3.1.1.5 Personalization. Modifications to the baseline cabinet assembly, NAVSEA drawing 53711-5942190, are permissible to meet specific program requirements. All personalizations must meet all requirements specified herein.

3.1.2 Rear connector panel. A rear connector panel shall provide for all electrical inputs/outputs (I/Os) at the rear of the enclosure. The rear connector panel shall mount to the rear structure. The rear connector panel may be flat as shown on

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figure 2 or it may protrude as shown on figure 1, provided that the required cabling and water connections can be accomplished within the balance of the 12 inch (304.8 mm) access area of 3.15.5.

3.1.3 SEM drawers. SEM drawer designs shall be provided to accommodate various SEM electronic modules, in accordance with MIL-STD-1389 and MIL-M-28787; SES backplane assemblies in accordance with MIL-STD-2119 and MIL-A-28870 for printed wiring backplane assemblies and MIL-STD-2198 and MIL-A-24741 for metal backplane assemblies; and SES backplane components and support hardware in accordance with MIL-C-28754 and MIL-C-28859. Descriptions and sources of various SEM and SES hardware items that can be utilized in SEM drawers are provided in TP-532. Procedures for implementation of SEM are specified in MIL-STD-1378, and procedures for implementation of SES hardware are specified in MIL-STD-2200. Every SEM drawer shall be constructed to a common form factor with respect to its interface with the front structure, rear structure, slide assemblies, and the water manifold. Each drawer shall be capable of housing two power supplies and providing for a density equivalent of not less than 240 Format B 2A module slots or 120 Format D module slots. The drawer front panel shall have approximate dimensions of 7.5 inches (190.5 mm) wide by 23.2 inches (589.3 mm) high and shall not exceed 3.5 inches (88.9 mm) deep with power supplies attached. A mounting interface with adequate cooling capacity shall be provided by the front panel to accommodate two SPS power supplies, family S1, style 3 as specified in MIL-P-24764, having dimensions not to exceed 6.83 inches (173.5 mm) wide by 10.80 inches (274.3 mm) high by 2.51 inches (63.8 mm) deep. Procedures for the implementation of SPS are specified in MIL-STD-2038. The portion of the drawer housed within the enclosure shall be capable of passage through the front structure openings having maximum dimensions of 6.7 inches (170.2 mm) wide by 22.75 inches (577.9 mm) high. The drawer depth shall be a nominal 15 inches (381 mm) beyond the surface of the front panel. The weight of a fully populated drawer in its operational configuration shall not exceed 200 pounds (90.72 kilograms (kg)). A baseline Format D drawer has been developed, and its configuration is documented in NAVSEA drawing 53711-5942149.

3.1.3.1 SEM drawer performance characteristics. Each SEM drawer shall provide sufficient cooling capacity to insure all installed electronic components meet or exceed their design reliability goals. The drawer cooling system shall be capable of maintaining SEM module rib temperatures and SPS power supply baseplate temperatures below 140°F (60°C) when supplied with cooling water at temperatures up to and including 95°F (35°C) on submarines or 105°F (40.6°C) on surface ships with the SEM modules at maximum recommended power dissipation per MIL-STD-1389

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and SPS power supplies at maximum power dissipation per MIL-P-24764. Cooling water will be provided via the cabinet assembly (NAVSEA drawing 53711-5942190) water manifold system. The inlet/outlet fittings of the drawer shall be compatible with the hose fittings of the cabinet assembly water manifold system. The flow velocity within the cooling system of the drawer shall not exceed 15 feet per second (4.57 meters per second), and the total pressure drop for each drawer shall not exceed 8.5 pounds per square inch (psi) (58.62 kilopascal (kPa)). Each drawer shall include adequate cooling capacity to dissipate 2 kilowatts (KW) uniformly distributed. All cooling water passages, including connections, shall be capable of withstanding normal operating pressures up to 205 pounds per square inch gage (psig) (1.52 megapascal (MPa)) and a proof pressure of 275 psig (2 MPa) without leakage, damage or operational degradation when tested in accordance with 4.9.1. The SEM drawer shall survive, without damage or performance degradation, the environmental requirements of 3.20 when tested in a CE^3.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.5.

3.3 Modularity. The enclosure shall be constructed in a modular fashion that will enable loading of dismantled enclosure components into interior equipment spaces through hatches, bulkhead doors, and other passageways for reassembly and installation on the deck foundation. The enclosure design shall facilitate such disassembly/reassembly in order to minimize the onboard effort. The disassembly/reassembly shall be accomplished with standard, readily available tools and lifting aids. The use of coordinated tooling and self jigging shall be employed to assure repeatability and interchangeability. When their use is approved by the procuring activity, matched set items shall be documented in installation instructions as well as in engineering drawings in accordance with DOD-STD-100. Overall dimensions and their worst case tolerances of knocked down enclosure subassemblies shall be compatible with minimum hatch, door, and passageway sizes. All seams and joints shall be sealed against electromagnetic interference (EMI) and moisture in accordance with MIL-STD-2036.

3.4 Size and weight. The enclosure shall be a maximum of 72 inches (1829 mm) high with a maximum footprint of 24 inches (610 mm) wide by 24 inches (610 mm) deep, excluding any removable lifting aids (such as eye bolts). No protrusions shall be allowed forward of the 24 inch (610 mm) footprint. The maximum I/O depth available at the enclosure rear is 12 inches (305 mm), and the access space in front of the enclosure will be 21 inches (533 mm) (see 3.15.5). The gross weight of the fully populated

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enclosure in its operational configuration shall not exceed 2600 pounds (1179 kg).

3.5 CE^3 performance characteristics. The CE^3 shall provide sufficient cooling capacity to insure all installed electronic components meet or exceed their design reliability goals. Except as modified herein, the water cooling system of the CE^3 shall be in accordance with MIL-W-21965, cooling method 2. The enclosure cooling system shall be capable of maintaining SEM module rib temperatures and SPS power supply baseplate temperatures below 140°F (60°C) when supplied with cooling water at temperatures up to and including 95°F (35°C) on submarines or 105°F (40.6°C) on surface ships with the SEM modules at maximum recommended power dissipation per MIL-STD-1389 and SPS power supplies at maximum power dissipation per MIL-P-24764. Cooling water will be provided as per 3.15.4. The flow velocity within the cooling system of the enclosure shall not exceed 15 feet per second (4.57 meters per second), and the total pressure drop for the enclosure shall not exceed 10 psi (68.97 kPa). The CE^3 cooling capacity shall provide for dissipation of up to and including 18 KW. All cooling water passages, including connections, shall be capable of withstanding normal operating pressures up to 205 psig (1.52 MPa) and a proof pressure of 275 psig (2 MPa) without leakage, damage or operational degradation when tested in accordance with 4.9.1.

3.6 Parts, materials, and processes. The parts, materials, and processes selected for construction of the CE^3 shall be in accordance with MIL-STD-2036. Additionally, the parts, materials, and processes for the water cooling system shall conform to MIL-W-21965.

3.7 Electromagnetic radiation. Electromagnetic Interference/Electromagnetic Compatibility (EMI/EMC) requirements for the CE^3 shall be determined as part of the individual equipment specification and shall be in accordance with MIL-STD-461.

3.8 Workmanship. Workmanship shall be in accordance with requirement 9 of MIL-STD-454.

3.9 Interchangeability. Interchangeability shall conform to requirement 7 of MIL-STD-454.

3.10 Safety. Safety requirements shall comply with requirement 1 of MIL-STD-454 and ASTM F 1166.

3.11 Human engineering. Human engineering design criteria and principles shall conform to the requirements of ASTM F 1166.

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3.12 Grounding, bonding, and shielding. Electrical grounding, bonding, and shielding of the CE^3 shall be in accordance with MIL-STD-2036 and MIL-STD-1310.

3.13 Marking. Product marking shall be incorporated into the CE^3 in accordance with requirement 67 of MIL-STD-454.

3.14 Handling. Any components of the CE^3 weighing more than 80 pounds, ready for passage through a hatch, shall be provided with lifting provisions. Removal of lifting eyebolts from the CE^3 shall not affect the dripproof requirements of 3.20.10 herein.

3.15 Installation interfaces. The CE^3 shall interface with and be compatible with the ship and other combat system equipments.

3.15.1 Foundation. The CE^3 mechanical interface with the ship shall be at a deck foundation only. The mechanical interface shall facilitate enclosure installation/removal with minimum system disassembly and without complicated reassembly procedures. Foundation details (such as flatness and stiffness) required for enclosure system installation shall be provided in installation control drawings which shall be the responsibility of the contractor.

3.15.2 Electrical power. Electrical power shall be provided to the CE^3 in accordance with MIL-STD-1399, section 300. The electrical power system characteristics provided shall be governed by the individual specification requirements for the electronic equipment to be installed in the particular enclosure.

3.15.3 Electrical/electronic signal cabling. Electrical interface with the electrical power system and the electronic signal interface between enclosures shall be provided by connectors and mating receptacles located on the enclosure rear connector panel. All interface connectors shall be in accordance with MIL-C-5015 or MIL-C-28840.

3.15.4 Cooling water. The cooling water system of the CE^3 shall conform to the interface requirements and constraints of DOD-STD-1399, section 532. The CE^3 shall be provided with cooling water from the ship's EAFW system via a single supply connection and a single return connection at the upper rear of the enclosure. The supply and return interface connections shall be compatible with the threaded fittings of NAVSEA drawing 53711-5942198. The maximum temperature of the EAFW supply to the enclosure is 95°F (35°C) on submarines and 105°F (40.6°C) on surface ships.

3.15.5 Access area. An unobstructed, 21 inch (533.4 mm)

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deep service space will be provided in front of the enclosure starting at the forward edge of the footprint as shown on figure 3. In general, the 21 inch (533.4 mm) deep service space will extend a minimum of 18 inches (457.2 mm) to either side of the enclosure front; but, in those cases where the enclosure is to be located adjacent to a bulkhead or other obstruction, the enclosure drawer design shall provide for orientation of the SEM modules such that they are accessible for installation/removal from the side of the drawer away from the obstruction. A cabling and water connection access and routing space will be provided behind the enclosure starting at the rear edge of the footprint and extending 12 inches (304.8 mm) beyond. This space will provide for routing of waterlines, cables and all I/O connections (including connector panels) over the enclosure width, extending vertically from the top rear enclosure edge downward to within 15 inches (381 mm) of the deck. In some cases, the lower 15 inch (381 mm) space may be occupied by a ships service air plenum, and therefore may only be available to accommodate rear protrusions on an individual case basis.

3.16 Reliability. The CE^3 will be primarily concerned with enhancing the reliability of its installed electronics equipment. Therefore, the enclosure system reliability shall be emphasized in the mechanical design of the components required to perform the desired functions (structural support and protection from environmental conditions). Emphasis shall also be placed on system reliability in the areas of hardware selection and utilization; including slides, assembly hardware, hoses, valves, seals, fasteners, and wiring harnesses.

3.16.1 Reliability requirements. The CE^3 shall be designed to perform its primary function as an environmental enclosure with a 95 percent confidence level for a period of not less than twenty years, assuming normal maintenance and replacement of degradable parts such as hoses and gaskets. All movable parts, which are actuated for inspection or maintenance, such as latches, actuators, lock devices, drawer slide assemblies, front panels, doors, hinges, water hoses/distribution assemblies, and wiring harnesses shall be designed for a minimum life of 1000 cycles.

3.17 Maintainability. The CE^3 design shall provide for and support expedient access for maintenance of all modules and power supplies without the necessity of removing the drawer from the enclosure for any installed location. Additionally, any maintenance action required on a particular drawer, including drawer removal/replacement, shall be possible without affecting any other drawer. Furthermore, all CE^3 maintenance shall be capable of being performed with standard, readily available tools.

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3.17.1 Maintainability requirements. The CE^3 maintainability requirements are key elements in the specified user system Mean Time To Repair (MTTR). As a minimum, the following maintenance requirements for a fully populated, operational CE^3 shall be met.

3.17.1.1 SEM module replacement. SEM module replacement time shall be specified and demonstrated at 5.3 minutes or less. Replacement time excludes the electronic diagnostic time to isolate the module to be replaced. Replacement time includes time required to identify drawer containing module; time to unsecure and open drawer; time to identify specified module; time to remove and replace module; and time to close drawer and secure.

3.17.1.2 SPS power supply replacement. SPS power supply replacement time shall be specified and demonstrated at 4.0 minutes or less. Replacement time excludes the electronic diagnostic time to isolate the power supply to be replaced. Replacement time includes time required to identify specified power supply; time required to unsecure and remove power supply; and time required to replace and secure power supply.

3.17.1.3 Drawer replacement. Removal and replacement time requirements for any drawer of the CE^3, including verification of proper installation and operation, shall be demonstrated at 2.0 hours or less.

3.17.1.4 CE^3 removal and replacement. The CE^3 shall be designed such that removal and replacement of the enclosure can be accomplished within a 24 hour period. Removal and replacement of a CE^3 shall include:

(a) Disconnection of the existing CE^3 from its installed location within the submarine or ship.

(b) Disassembly (for transporting) and transportation out through hatches, doors, and passageways.

(c) Disassembly and transportation of the replacement CE^3 to the required location within the submarine or ship.

(d) Reassembly and reconnection of the replacement CE^3.

(e) Testing to verify proper installation and operation of the replacement CE^3.

Removal and replacement, excluding electronic verification testing, shall be demonstrated at 24 hours or less, contingent upon lack of other physical shipboard obstructions.

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3.18 Drain holes. The enclosure shall be provided with drain holes positioned to preclude the entrapment or retention of any fluid within the enclosure.

3.19 Noise requirements.

3.19.1 Structureborne noise. The CE[3] shall meet the structureborne noise requirements of MIL-STD-740-2 for type III equipment under conditions specified in the individual equipment specification when tested as specified in 4.9.3.

3.19.2 Airborne noise. The CE[3] shall meet the airborne noise requirements of MIL-STD-740-1 for grade C equipment under conditions specified in the individual equipment specification when tested as specified in 4.9.4.

3.20 Environmental requirements.

3.20.1 Low temperature (storage). The CE[3] shall survive the low temperature storage testing specified in 4.9.5 without any binding of moving parts and with no loss of resilience or sealing ability of seals, gaskets, hoses, and any other rubber or rubberlike items upon return to normal operating temperature range. No permanent distortion of any item is permitted.

3.20.2 High temperature (storage). The CE[3] shall survive the high temperature storage testing specified in 4.9.6 without any binding of moving parts and with no loss of resilience or sealing ability of seals, gaskets, hoses, and any other rubber or rubberlike items upon return to normal operating temperature range. No permanent distortion of any item is permitted.

3.20.3 Low temperature (operation). The CE[3] shall survive the low temperature operation testing specified in 4.9.7 without any binding of moving parts and with no loss of resilience or sealing ability of seals, gaskets, hoses, and any other rubber or rubberlike items upon return to normal operating temperature range. No permanent distortion of any item is permitted. An operational checkout during the low temperature operation testing shall show no electrical/electronic degradation of the equipment. No leakage of water is permitted.

3.20.4 High temperature (operation). The CE[3] shall survive the high temperature operation testing specified in 4.9.8 without any binding of moving parts and with no loss of resilience or sealing ability of seals, gaskets, hoses, and any other rubber or rubberlike items upon return to normal operating temperature range. No permanent distortion of any item is permitted. An operational checkout during the high temperature operation

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testing shall show no electrical/electronic degradation of the equipment. No leakage of water is permitted.

3.20.5 Humidity. The CE^3 shall withstand the humidity test as specified in 4.9.9 with no change in mechanical properties, no absorption of water and swelling, no internal corrosion, no form of deterioration, and no condensation detrimental to the electrical/electronic functioning of the equipment.

3.20.6 Salt fog. The CE^3 shall withstand the salt fog test as specified in 4.9.10. The finish of the samples tested shall not show any evidence of blistering, softening, separation from the base metal, or corrosion products (for example, aluminum oxide and iron oxide).

3.20.7 Vibration. The CE^3 shall be tested as specified in 4.9.11. The first mode vibration frequency shall be sufficiently greater than 33 Hz to insure that amplification of base-applied vibratory excitations shall not exceed 3.0 to 1.0 at any point on the front structure and rear structure over a frequency range from 4 to 33 Hz. Amplification of base applied vibratory excitations shall not exceed 4.0 to 1.0 anywhere within the CE^3 over a frequency range from 4 to 33 Hz. Moreover, there shall be no failure, significant structural deformation, or yielding of any parts.

3.20.8 Reduced level vibratory scan. The CE^3 shall be subjected to the reduced level vibratory scan test specified in 4.9.12 prior to and following the shock tests of 4.9.13. There shall be no change in the structural response measured during the vibratory scan test performed after the testing of 4.9.13 as compared with the vibratory scan performed prior to the testing of 4.9.13.

3.20.9 Shock. The CE^3 shall be tested as specified in 4.9.13. There shall be no failure, significant structural deformation, or yielding of any parts. Following the shock testing, the vibratory scan test of 4.9.12 shall be performed to check for damage. There shall not be any combination of external or internal environmental stresses that contribute to an increase of the CE^3 structural response(s) to a level in excess of those specified herein.

3.20.10 Dripproof. The CE^3 shall meet the dripproof requirements of MIL-STD-108 when tested as specified in 4.9.14.

3.20.11 Inclination. The CE^3 shall withstand the inclination test specified in 4.9.15 with no permanent deformation of material and no binding of slides or any other moving parts. Normal electrical/electronic functioning of the

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equipment shall be verified after each of the cyclic tests performed per 4.9.15.

3.20.12 Low pressure (altitude). The CE[3] shall withstand the low pressure (altitude) testing specified in 4.9.16 with no physical or performance degradation.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.5).

(b) Quality conformance inspection (see 4.6).

4.3 Inspection conditions. Unless otherwise specified in the contract or test procedures of this specification, inspections shall be performed in an area having a temperature of 25°C ± 5°C (77°F ± 9°F) and a relative humidity of up to 90 percent. The barometric pressure shall be between 24 and 32 inches (610 and 813 mm) of mercury.

4.4 Test plans and procedures. The contractor shall provide test plans and test procedures for all test categories in accordance with the applicable procurement documents. Test plans and procedures shall be submitted to the government procuring activity for review and approval prior to beginning inspection. Subsequent changes or revisions to approved test procedures shall be subject to government approval.

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4.5 First article inspection. Unless otherwise specified in the contract (see 6.2), first article tests shall be performed on one enclosure representing the enclosures to be supplied under the contract. The inspection shall consist of the examinations and tests as specified in table I, in the order shown.

4.5.1 First article inspection report. The contractor shall furnish a first article inspection report in accordance with the data ordering document specified in 6.3.

TABLE I. First article inspection.

Inspection	Requirement paragraph	Method paragraph
Visual and workmanship	3.1, 3.6, 3.8	4.8
Dimensions and mechanical	3.1, 3.3, 3.4	4.7
Low temperature (storage)	3.20.1	4.9.5
High temperature (storage)	3.20.2	4.9.6
Low temperature (operation)	3.20.3	4.9.7
High temperature (operation)	3.20.4	4.9.8
Humidity	3.20.5	4.9.9
Salt fog	3.20.6	4.9.10
Vibration	3.20.7	4.9.11
Vibratory scan	3.20.8	4.9.12
Shock	3.20.9	4.9.13
Dripproof	3.20.10	4.9.14
Inclination	3.20.11	4.9.15
Low pressure (altitude)	3.20.12	4.9.16
Hydrostatic pressure	3.5	4.9.1
Mechanical life cycling	3.16.1	4.9.2
Structureborne noise	3.19.1	4.9.3
Airborne noise	3.19.2	4.9.4

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspection.

4.6.1.1 Group A inspection. Group A inspection shall be conducted by the contractor and shall consist of the inspections specified in table II. The procuring activity reserves the right to inspect the facilities, equipment, and procedures used in the group A inspection.

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TABLE II. Group A inspection.

Inspection	Requirement paragraph	Method paragraph
Visual and workmanship	3.1, 3.6, 3.8	4.8

4.6.1.2 Group B inspection. Group B inspection shall be conducted by the contractor and shall consist of the inspections specified in table III. The procuring activity reserves the right to inspect the facilities, equipment and procedures used in the group B inspection.

TABLE III. Group B inspection.

Inspection	Requirement paragraph	Method paragraph
Dimensions and mechanical	3.1, 3.3, 3.4	4.7

4.6.2 Periodic inspection. Unless otherwise specified in the solicitation or acquisition documents, periodic inspection shall be performed as follows. Periodic inspection shall consist of group C inspections specified herein. Where results of these inspections show noncompliance with the applicable requirements (see 4.6.2.1.1), the delivery of products that have passed groups A and B shall not be delayed pending analysis of the results of these periodic inspections.

4.6.2.1 Group C inspection. Group C inspections shall consist of the inspections specified in table IV, in the order shown. Group C inspection shall be performed on an enclosure that has passed groups A and B inspections. The group C candidate enclosure shall not have been screened and may be

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selected by a representative of the procuring activity. The procuring activity reserves the right to inspect the facilities, equipment and procedures used in the group C inspection.

4.6.2.1.1 Noncompliance. If an enclosure fails to pass group C inspections and tests, the manufacturer shall notify the procuring activity and the cognizant inspection activity of such failure. The manufacturer shall take corrective action on the materials and process, as warranted, and on all items which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials and processes, and which are considered to be subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action acceptable to the procuring and cognizant inspection activities has been taken. After corrective action has been taken, group C inspection shall be repeated, either all inspections or the inspection which the sample enclosure failed, at the option of the procuring activity. Groups A and B may be reinstituted; however, final acceptance and shipment shall be withheld until group C inspections have shown that the corrective action was successful. In the event of failure after reinspection, the manufacturer shall notify the procuring and cognizant inspection activities.

TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Method paragraph
Low temperature (storage)	3.20.1	4.9.5
High temperature (storage)	3.20.2	4.9.6
Low temperature (operation)	3.20.3	4.9.7
High temperature (operation)	3.20.4	4.9.8
Humidity	3.20.5	4.9.9
Salt fog	3.20.6	4.9.10
Vibration	3.20.7	4.9.11
Vibratory scan	3.20.8	4.9.12
Shock	3.20.9	4.9.13
Dripproof	3.20.10	4.9.14
Inclination	3.20.11	4.9.15
Low pressure (altitude)	3.20.12	4.9.16
Hydrostatic pressure	3.5	4.9.1
Mechanical life cycling	3.16.1	4.9.2
Structureborne noise	3.19.1	4.9.3
Airborne noise	3.19.2	4.9.4

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4.6.3 Quality conformance inspection report. The contractor shall furnish a quality conformance inspection report in accordance with the data ordering document specified in 6.3.

4.7 Dimensions and weights. Dimensions and weights of all units comprising the complete equipment and all the repair parts shall be determined and shall conform to the requirements of 3.1, 3.3, and 3.4.

4.8 General examination. The completed equipment shall be given a thorough examination for the configuration requirements of 3.1; the parts, materials, and processes requirements of 3.6; and the workmanship requirements of 3.8.

4.9 Tests.

4.9.1 Hydrostatic pressure. The water cooling system of the CE^3 shall be subjected to the hydrostatic pressure test of MIL-W-21965, except test pressure shall be 275 psig (2 MPa). The water cooling system shall meet the requirements of 3.5.

4.9.2 Mechanical life cycling. Life cycle tests shall be conducted to demonstrate that latches, actuators, lock devices, drawer slide assemblies, front panels, door, hinges, water hoses/distribution assemblies, electrical connections, wiring harnesses, and other movable parts meet the cycling requirements of 3.16.1. Existing designs may be qualified by similarity, provided adequate and accurate reliability data is provided.

4.9.3 Structureborne noise. A fully populated, operational CE^3 shall be tested in accordance with MIL-STD-740-2 to meet the structureborne noise requirements of 3.19.1.

4.9.4 Airborne noise. A fully populated, operational CE^3 shall be tested in accordance with MIL-STD-740-1 to meet the airborne noise requirements of 3.19.2.

4.9.5 Low temperature (storage). The low temperature test shall be performed in accordance with MIL-STD-810, method 502.3, procedure I (storage). The test shall be conducted on a fully populated, operational CE^3, purged of cooling water, at a test temperature of minus 40°C (minus 40°F). After the unit is stabilized at minus 40°C (minus 40°F), the temperature shall be maintained for a minimum of 4 hours. Following the low temperature storage test, the CE^3 shall meet the requirements of 3.20.1.

4.9.6 High temperature (storage). The high temperature test shall be performed in accordance with MIL-STD-810, method 501.3, procedure I (storage). The test shall be conducted on a fully

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populated, operational CE^3, purged of cooling water. The test shall consist of a minimum of seven temperature cycles of 24 hours duration based on climatic category A1 (hot). The maximum temperature of 70°C (158°F) shall be held for approximately one hour in each cycle. Following the high temperature storage test, the CE^3 shall meet the requirements of 3.20.2.

4.9.7 Low temperature (operation). The low temperature test shall be performed in accordance with MIL-STD-810, method 502.3, procedure II (operation). The test shall be conducted on a fully populated, operational CE^3. With the CE^3 energized and in the test chamber, adjust the chamber air temperature to 0°C (32°F). Maintain until temperature stabilization of the CE^3 has been achieved. During and following the low temperature operation test, the CE^3 shall meet the requirements of 3.20.3.

4.9.8 High temperature (operation). The high temperature test shall be performed in accordance with MIL-STD-810, method 501.3, procedure II (operation). The test shall be conducted on a fully populated, operational CE^3. The test shall consist of seven temperature cycles of 24 hours duration. The maximum temperature shall be 50°C (122°F), and it shall be held for approximately one hour in each cycle. The CE^3 shall be operated during the maximum response period of the exposure cycle. During and following the high temperature operation test, the CE^3 shall meet the requirements of 3.20.4.

4.9.9 Humidity. The humidity test shall be performed in accordance with MIL-STD-810, method 507.3, procedure III. The test shall be conducted on a fully populated, operational CE^3. Following the humidity test, the CE^3 shall meet the requirements of 3.20.5.

4.9.10 Salt fog. The salt fog test shall be conducted in accordance with MIL-STD-810, method 509.3, procedure I on selected parts and finishes. Those parts and finishes to be tested shall be identified in the approved detailed first article inspection procedures (see 6.4). It will not be applied to complete sets or units. Sample corner structures and any other critical sections may be used for the test. The sample shall be assembled and treated in the same manner as the complete structure. Following the salt fog test, the test samples shall meet the requirement of 3.20.6.

4.9.11 Vibration. A fully populated CE^3 shall be tested in accordance with MIL-STD-167-1, type I vibration requirements, over a frequency range of 4 to 33 Hz. The CE^3 shall meet the requirements of 3.20.7.

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4.9.12 Reduced level vibratory scan. The reduced level vibratory scan tests shall be performed on a fully populated CE^3 in accordance with the exploratory vibration test of MIL-STD-167-1. The vibratory level (amplitude) shall be as prescribed in MIL-STD-167-1, except as otherwise specified in the contract or purchase order. Following the vibratory scan tests, the CE^3 shall meet the requirements of 3.20.8.

4.9.13 Shock. A fully populated CE^3 shall be tested in accordance with MIL-S-901 for grade A, deck mounted, class I, medium weight, type A shock requirements. The CE^3 shall be mounted on a soft deck simulator in accordance with drawing 88594-01257. Nine shock blows shall be applied in a sequence consisting of three vertical, followed by three front over 30° and concluded with three side over 30°. The CE^3 shall be energized to perform its normal function during the shock tests. Following the shock tests, the CE^3 shall meet the requirements of 3.20.9.

4.9.14 Dripproof. The drippoof test shall be conducted in accordance with MIL-STD-108. The CE^3 shall meet the dripproof requirements of 3.20.10.

4.9.15 Inclination. A fully populated operational CE^3 shall be subjected to the test limits specified herein. The CE^3 shall be energized and fully operating during the applicable test. The CE^3 shall meet the requirements of 3.20.11.

4.9.15.1 Inclination test limits for surface ships. The enclosure shall be inclined at the rate of 5 to 7 cycles per minute in one phase to angles of 45 degrees on both sides of the vertical for a minimum period of 30 minutes. During the inclination testing, the enclosure drawer exerting the greatest load on the drawer slides shall be in the locked fully open position. The test shall be repeated with the enclosure reoriented 90 degrees to the plane in which it was originally tested. At the conclusion of these cyclic tests, the cyclic motion shall be stopped and the inclination adjusted to an angle of 15 degrees. The enclosure shall then be operated for a sufficient period to ensure that the continuous operation can be maintained. The equipment shall then be rotated through the vertical to 15 degrees in the opposite direction, and the test for continuous operation shall be repeated. The test shall be repeated with the equipment reoriented 90 degrees to the plane in which it was originally tested.

4.9.15.2 Inclination test limits for submarines. Equipment intended for submarine installations shall be subjected to the test of 4.9.15.1 except that the maximum angle shall be 60 degrees.

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4.9.16 Low pressure (altitude). The low pressure (altitude) test shall be conducted in accordance with MIL-STD-810, method 500.3, procedure I under the following test conditions:

(a) Test altitude - 15,000 feet (4,570 meters).

(b) Altitude change rate - 2,000 feet/minute (10 meters/second).

(c) Duration of exposure - 1.0 hour minimum.

(d) Test item configuration - fully populated, operationally configurated, purged of cooling water.

Following the low pressure altitude test, the CE^3 shall meet the requirements of 3.20.12.

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5. PACKAGING

5.1 Enclosure preservation and packing. Preservation and packing shall be in accordance with level A, B, or C as specified in MIL-STD-2073-1, and as required by the contract or purchase order. Unless stated otherwise in the contract or purchase order (see 6.2), preservation compound shall not be used.

5.2 Marking. All shipping and storage containers shall be marked in accordance with the requirements of MIL-STD-129.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The enclosure covered by this specification is intended to accommodate Standard Electronic Modules and Standard Power Supplies for use as a freestanding cabinet in submarine interior equipment rooms and spaces, or in surface ship sheltered, interior equipment rooms. Maintenance on the end use electronic equipment is at the extended drawer level or from the exterior in the case of the front-mounted Standard Power Supplies. The end user, cognizant engineering activity, or contractor shall identify and be responsible for the selection and/or design of the electrical and electronic payload, including but not limited to the electronic modules, power supplies, wiring, and cabling.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Implementation of a parts control program, if required (see 3.6).
(c) Responsibility for inspection (see 4.1).
(d) Name and location of approved test laboratory (see 4.1).
(e) Approval of previously used test procedures, if required (see 4.4).
(f) Modification of previously used test procedures, if required (see 4.4).
(g) Selection of applicable levels of preservation, packing and packaging required (see 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.4	DI-NDTI-80808	Test Plans/Procedures	---
4.5.1, 4.6.2.1.1, 4.6.3	DI-RELI-80373	Equipment Inspection/ Testing Report	---

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations of the enclosure, approval of first article test results, and disposition of the first article enclosure. Solicitation or acquisition documents should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. If the enclosure is procured in conjunction with a specific system, the system's end use electronic payload (modules, power supplies, wiring, cables, etc.) should be installed and tested as a complete unit. The contracting officer must be specific in regard to the manner in which first article inspection is to be conducted.

6.5 Definitions. The following definitions are applicable to terms used herein.

6.5.1 Individual equipment specification. An individual equipment specification is the specification which defines the detailed characteristics and is used for acquisition of a particular equipment. (definition from MIL-E-16400H).

6.5.2 SES backplane assemblies. The assemblies which contain backplane components that are installed in specifically designed baseplates or printed wiring boards for mating and interconnecting SEM within SES structures. Design and quality requirements for SES backplane assemblies are specified in MIL-STD-2198 and MIL-A-24741, or MIL-STD-2119 and MIL-A-28870. (definition from MIL-STD-2200).

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6.5.3 SES backplane components. Connector components (such as female contacts, insulator housings, polarizing bushings, and grounding bushings), qualified within the SHARP program, which are installed in backplane assemblies for mating with and interconnecting SEM. SES backplane components are specified in MIL-C-28754 and MIL-C-28859. (definition from MIL-STD-2200).

6.5.4 SES support hardware. Qualified and non-qualified miscellaneous SES hardware designs (such as extender boards, module extractor tools, cable connector assemblies, and module retainers) which have been developed, tested, and sourced within the SHARP program for use in employing SEM within SES structures. SES support hardware designs are referenced in TP-532. (definition from MIL-STD-2200).

6.5.5 Standard electronic modules (SEM). Standardized functional circuit card assemblies which meet the design and quality requirements of MIL-STD-1389 and MIL-M-28787. (definition from MIL-STD-2200).

6.6.6 Standard power supplies (SPS). Families of standardized power supply assemblies which meet the design and quality requirements of MIL-P-24764 for shipboard use and MIL-P-29590 for airborne use. (definition from MIL-STD-2200).

6.6 Subject term (key word) listing.

SEM
SES
SHARP
SPS
Standard Electronic Modules
Standard Enclosure Systems
Standard Hardware Acquisition and Reliability Program
Standard Power Supplies

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activity:
DLA - GS

Preparing activity:
Navy - SH

Agent: NW

(Project 5975-1044)

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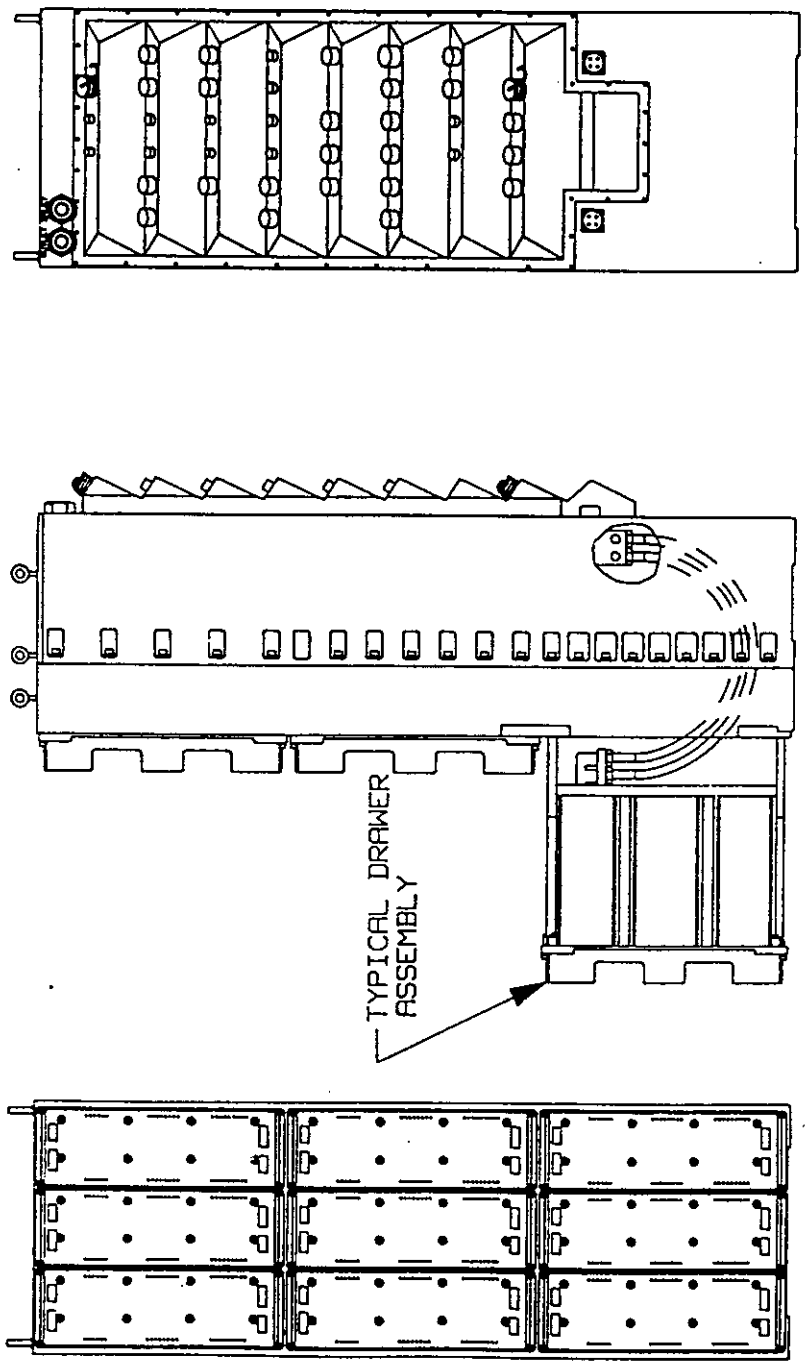


FIGURE 1. Typical CE[3].

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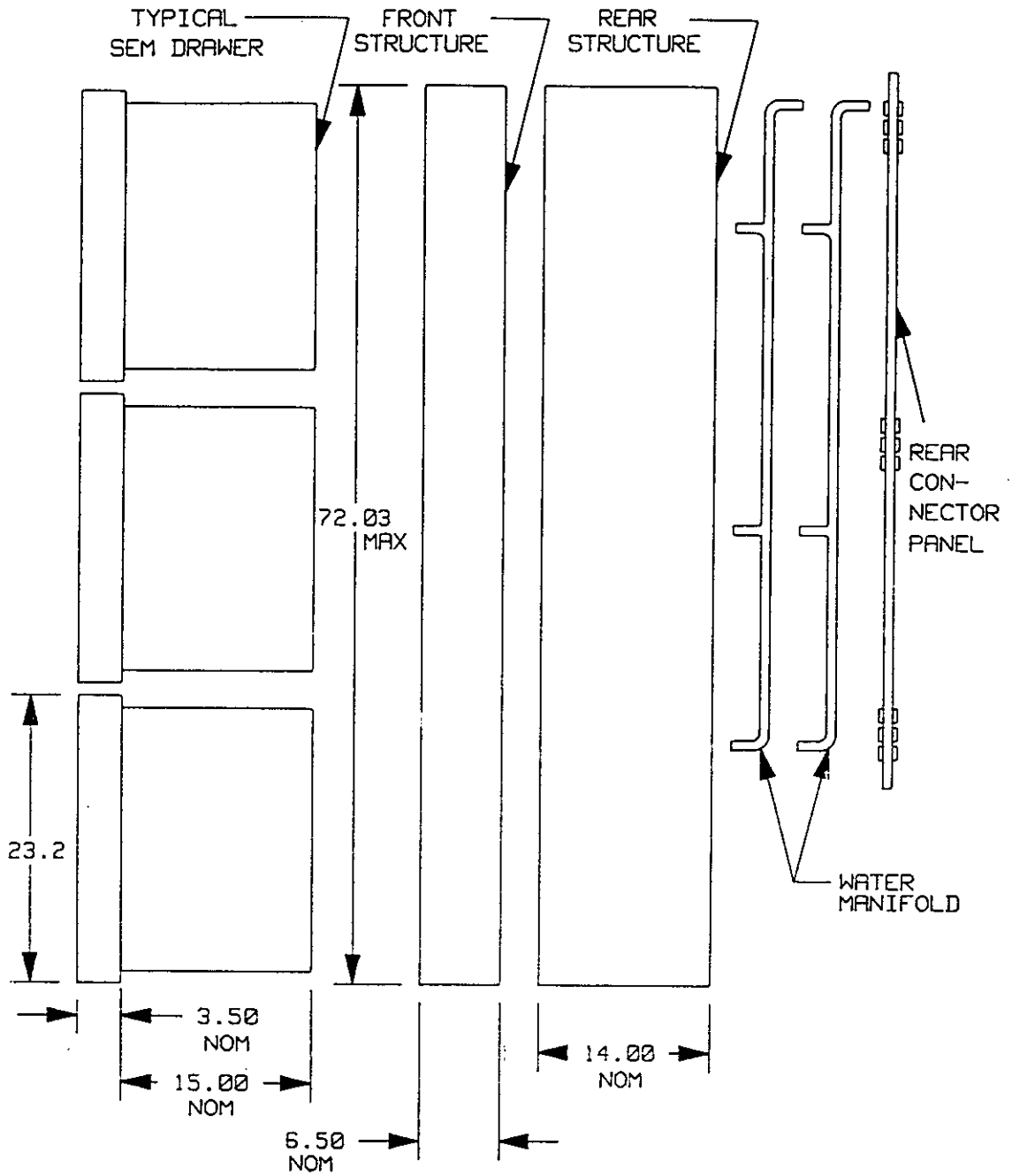


FIGURE 2. CE^3 major components.

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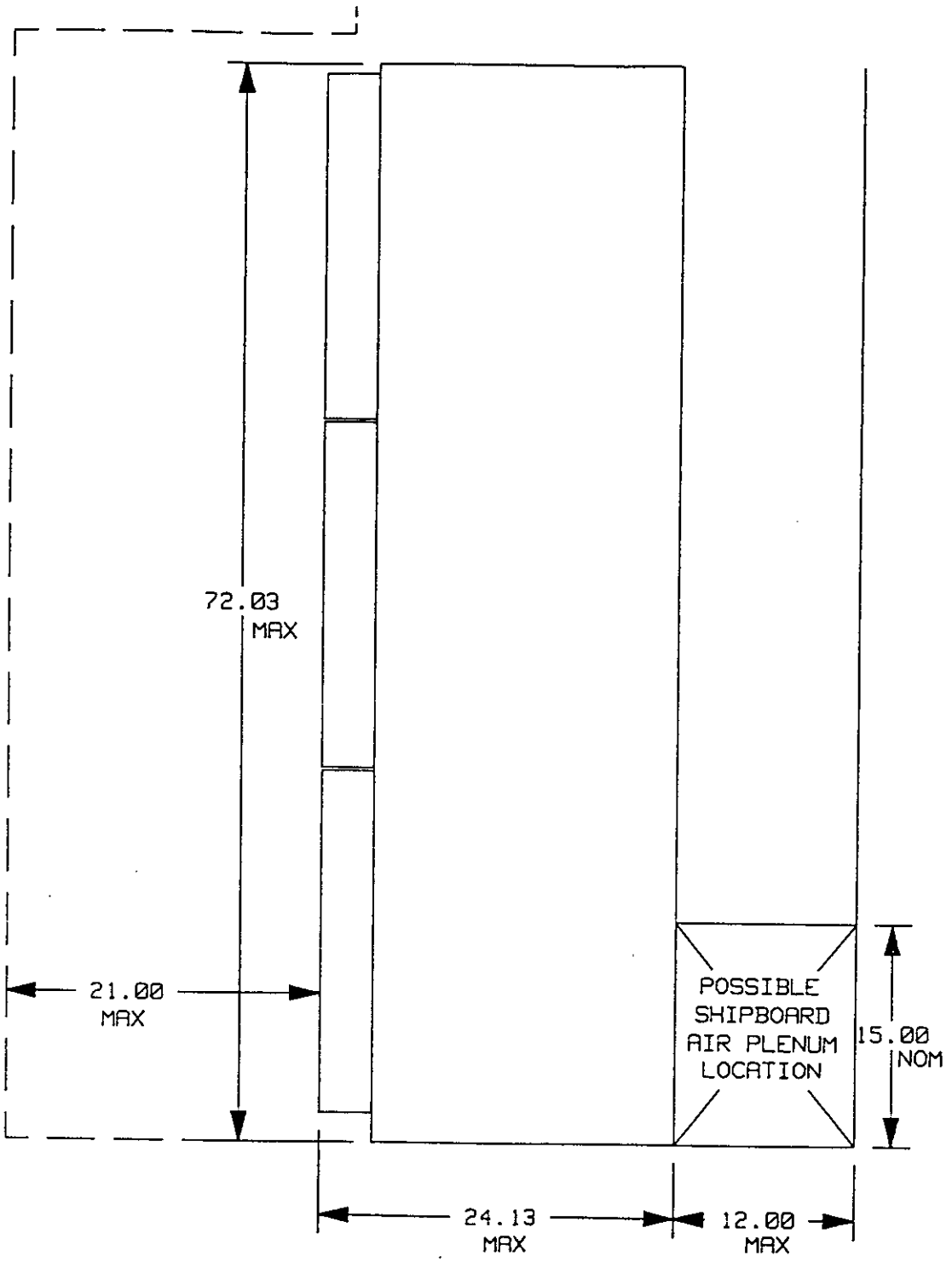


FIGURE 3. CE^3 access area.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-E-24743	

3. DOCUMENT TITLE
Enclosure, Common Electronic Equipment, General Specification for

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Technical Point of Contact (TPOC): Mr. Dan Geuder (Code 6043) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	TPOC: 812-854-3240	482-3240
c. ADDRESS (Include Zip Code) Commanding Officer Naval Weapons Support Center ATTN: Dan Geuder, Bldg. 2044, Code 6043 Crane, Indiana 47522-5060	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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